

BROWNWOOD, TX, USA

WMO: 722666

Lat: 31.800N

Lon: 98.950W

Elev: 1377

StdP: 13.98

Time Zone: -6.00 (NAC)

Period: 03-19

WBAN: 93943

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	20.7	24.9	7.4	8.5	31.7	11.7	10.5	37.8	26.0	51.9	23.7	56.5	4.2	0	0.464

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	23.9	100.3	71.7	99.2	72.0	97.0	72.6	77.9	90.4	76.8	89.3	75.7	87.8	9.7	150

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
74.8	137.4	81.3	73.2	130.1	80.5	72.7	127.5	80.1	42.3	90.8	41.3	89.9	40.3	86.0	83.8

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
22.0	19.1	17.3	DB	14.1	103.6	3.4	3.0	11.7	105.8	9.7	107.5	7.7	109.2	5.3	111.4	(4)
			WB	12.8	79.1	3.4	2.0	10.4	80.5	8.4	81.7	6.5	82.8	4.1	84.2	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	65.6	46.2	49.6	58.5	65.4	73.0	80.6	83.3	83.4	76.6	66.4	55.4	47.1
	DBStd	15.50	8.97	10.58	9.25	7.95	7.27	4.36	3.53	4.32	5.59	8.57	9.56	9.19
	HDD50	564	179	123	31	4	0	0	0	0	0	6	55	166
	HDD65	2319	585	437	238	90	19	0	0	0	1	86	307	556
	CDD50	6242	61	112	295	464	713	919	1031	1037	799	515	218	78
	CDD65	2522	2	7	37	101	267	469	566	572	350	129	20	2
	CDH74	30378	47	149	447	1221	2827	5570	7191	7355	3826	1424	256	65
	CDH80	15541	5	35	117	448	1268	2915	4068	4252	1859	510	55	9
Wind	WSAvg	7.5	7.0	8.0	8.7	9.4	8.6	8.2	7.0	6.3	6.0	6.6	7.1	6.8
Precipitation	PrecAvg	31.9	1.4	2.0	2.2	2.6	3.8	4.1	2.1	2.3	2.6	2.9	1.8	1.3
	PrecMax	45.2	3.9	7.3	6.8	10.0	10.4	12.0	6.5	9.8	5.4	6.5	6.4	8.0
	PrecMin	20.5	0.0	0.1	0.0	0.2	1.2	0.6	0.0	0.0	0.2	0.1	0.0	0.0
	PrecStd	7.0	1.0	1.9	1.7	2.1	2.1	2.7	1.8	2.6	1.7	2.2	1.9	1.7
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	80.8	85.6	88.0	95.5	99.2	101.7	103.6	104.0	99.6	92.8	85.9	81.2
		MCWB	58.3	61.1	62.4	64.4	71.7	70.9	70.1	72.3	69.8	69.6	65.8	59.9
	2%	DB	73.4	78.8	83.6	89.9	94.8	99.0	100.0	100.4	95.1	89.9	81.1	74.7
		MCWB	57.5	58.8	62.3	64.9	71.8	72.0	72.0	72.1	70.8	70.4	64.0	58.2
	5%	DB	69.6	73.3	79.3	84.3	90.8	95.4	98.9	99.4	92.7	85.6	76.6	69.9
		MCWB	56.5	57.6	61.9	64.6	70.9	72.3	72.5	72.2	71.3	68.0	61.7	56.2
	10%	DB	64.0	68.2	74.8	81.3	86.5	92.8	95.3	97.0	90.4	82.0	72.5	64.3
		MCWB	52.7	56.0	60.3	64.7	69.9	73.1	72.4	73.2	71.5	65.9	60.6	54.2
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	66.2	66.4	69.6	73.0	77.6	79.9	78.5	78.6	77.6	75.5	70.2	66.7
		MCDB	71.6	74.0	79.5	83.1	88.8	90.0	90.8	92.5	90.4	86.7	79.4	72.7
	2%	WB	62.6	64.0	67.1	70.8	75.5	77.9	77.3	77.4	75.7	73.2	67.8	62.8
		MCDB	67.9	71.1	75.7	80.5	86.2	89.2	90.4	91.2	86.7	81.8	74.6	68.9
	5%	WB	58.1	61.3	65.5	69.2	74.2	76.4	76.1	76.3	74.6	71.8	65.5	59.3
		MCDB	65.4	69.1	73.5	78.7	84.7	87.5	89.0	89.4	85.0	79.6	72.1	65.5
	10%	WB	54.0	58.1	63.7	67.6	72.7	75.0	75.2	75.1	73.4	70.2	63.3	55.6
		MCDB	62.7	66.6	70.5	75.7	83.1	85.9	87.8	87.9	83.3	77.7	69.2	63.6
Mean Daily Temperature Range	5% DB	MDBR	26.5	26.2	25.7	26.4	22.8	22.2	23.5	23.9	23.4	25.5	26.2	26.9
		MCDBR	35.6	35.3	31.3	32.6	29.0	26.3	27.4	27.4	27.1	28.2	31.4	35.8
		MCWBR	20.1	19.3	14.7	14.9	10.6	7.3	7.1	6.9	8.7	12.1	17.0	20.8
	5% WB	MCDBR	28.7	28.4	22.2	24.1	22.1	20.7	22.4	23.2	21.4	20.9	23.1	25.7
		MCWBR	19.3	18.4	12.4	13.3	9.6	8.3	7.3	7.4	8.1	10.5	15.0	17.9
Clear-Sky Solar Irradiance	taub	0.302	0.310	0.335	0.376	0.413	0.414	0.425	0.417	0.389	0.353	0.321	0.305	
	taud	2.521	2.505	2.449	2.357	2.313	2.347	2.366	2.371	2.393	2.481	2.500	2.519	
	Ebn at Noon	295	302	300	290	278	277	273	274	279	283	285	287	
	Edh at Noon	27	30	34	39	42	40	39	38	36	31	27	26	
All-Sky Solar Radiation	RadAvg	995	1200	1546	1869	1954	2192	2180	1998	1677	1342	1046	884	
	RadStd	106	170	125	158	193	174	158	137	137	202	132	113	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
(46)	Station Only	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(47)	Regional (36 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+207	N/A

Nomenclature: See separate page